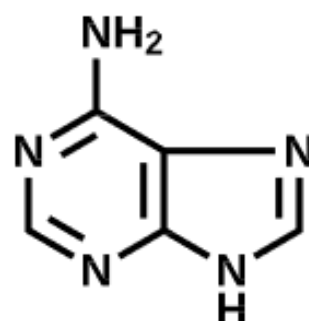
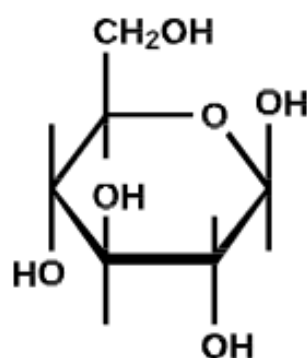
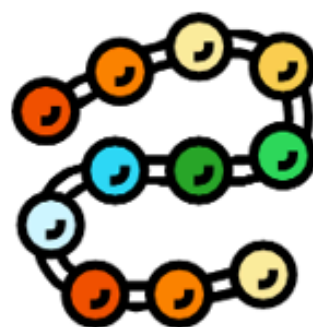
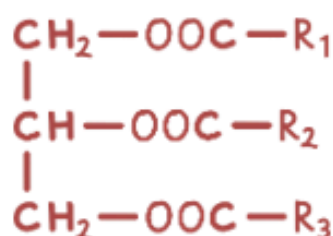


Unit 1:

Biochemistry

Name: _____



Topics:

Characteristics of Life

Graphing, Measurements, & Experimental Design

Properties of Water

Types of Reactions & Macromolecules

Name: _____

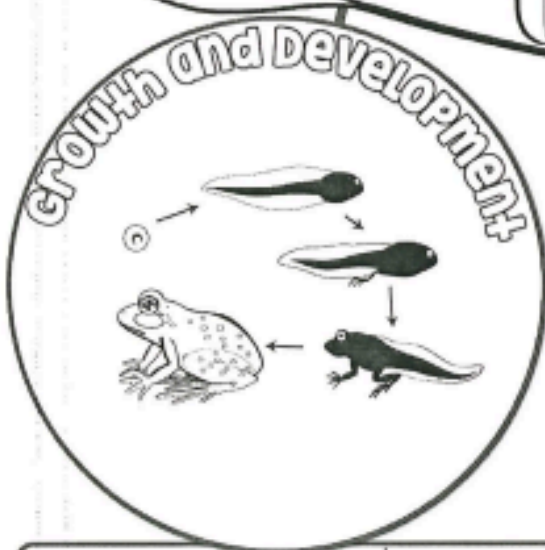
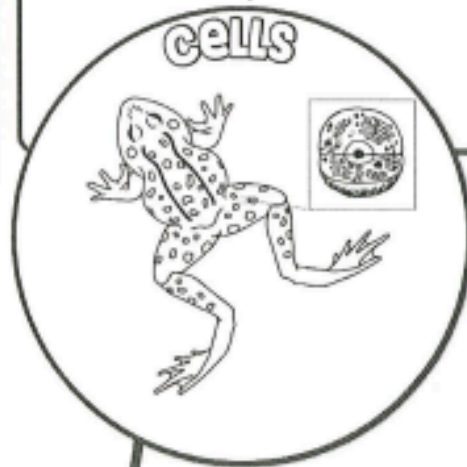
Autotrophs:

Heterotrophs:

Unicellular:

Multicellular:

Colonial Organism



Growth:

Development:

Sexual Reproduction:

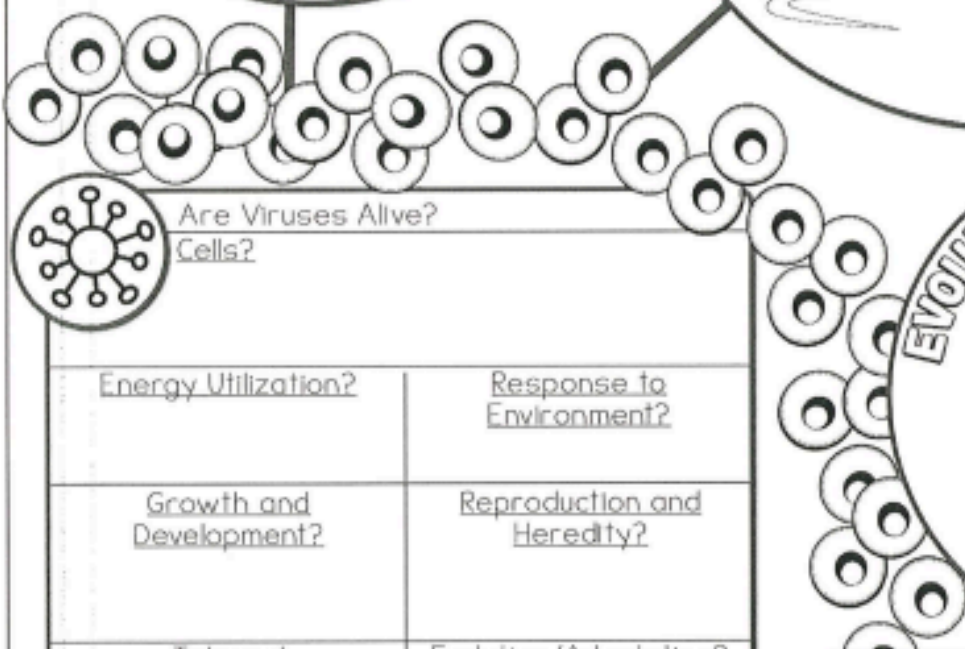
Asexual Reproduction:

Name: _____

Inside Cells:

Inside Organisms:

Internal regulation:



Are Viruses Alive?
Cells?

Energy Utilization?

Response to Environment?

Growth and Development?

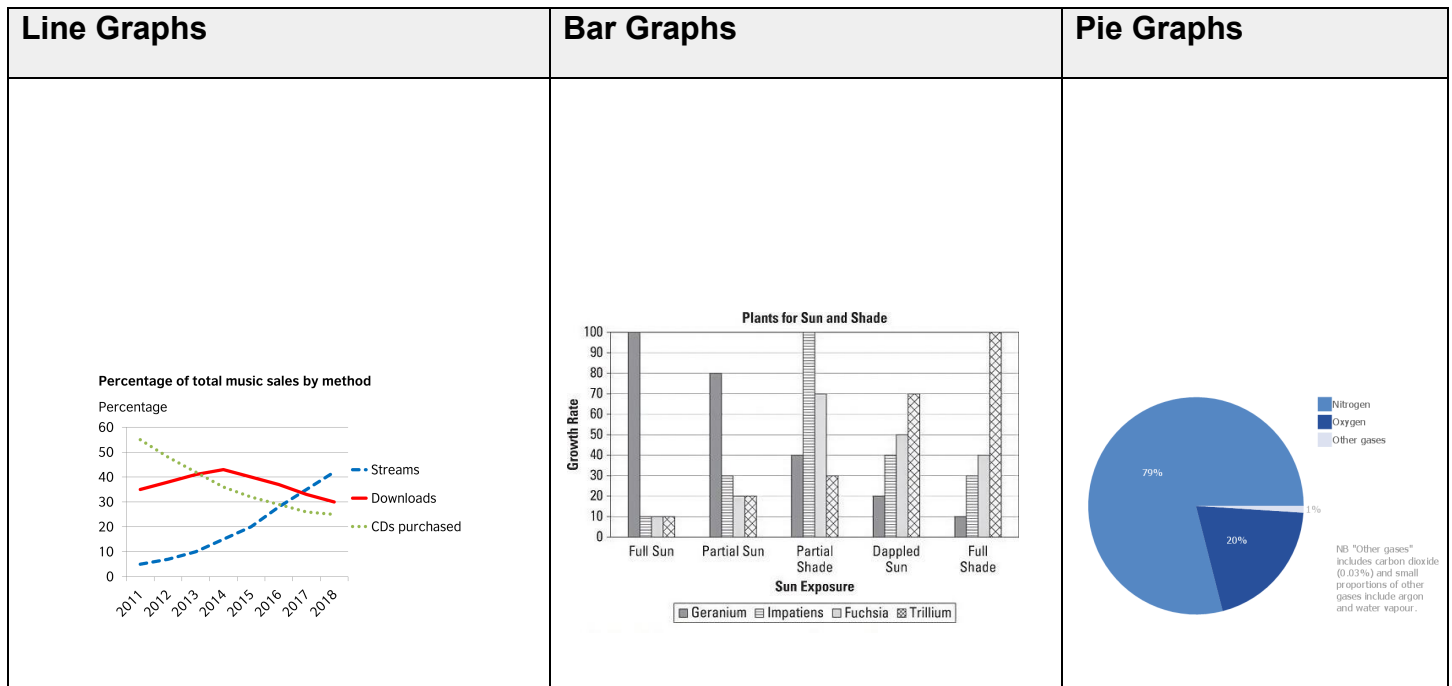
Reproduction and Heredity?

Internal Regulation/Order?

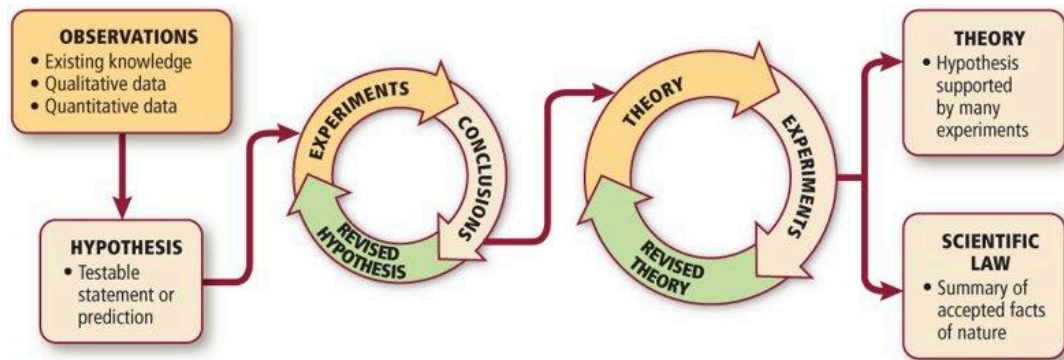
Evolution/Adaptation?



GRAPHING, MEASUREMENTS, & EXPERIMENTAL DESIGN



THE SCIENTIFIC METHOD



Is the scientific method considered a linear process? Why or why not?

Experimental Design Example

I want to run an experiment testing how caffeine affects the ability of grizzly bears to solve puzzles. There are four groups of bears. Group A gets no caffeine, Group B gets 100 mg of caffeine, Group C gets 200 mg of caffeine, and group D gets 400 mg of caffeine. The bears then all solve the same puzzle and the time it takes them to solve it is measured. Identify the following:

Control Group:

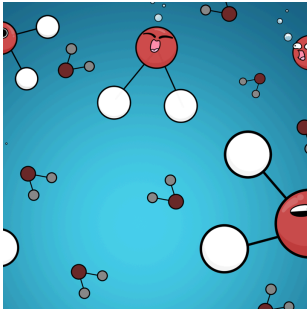
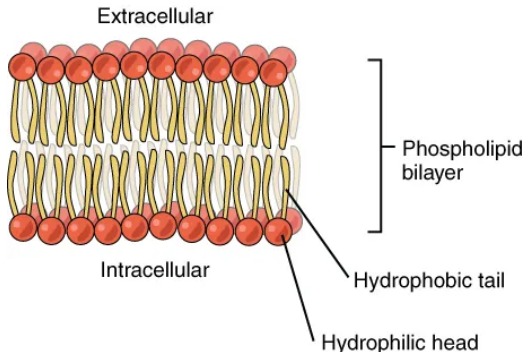
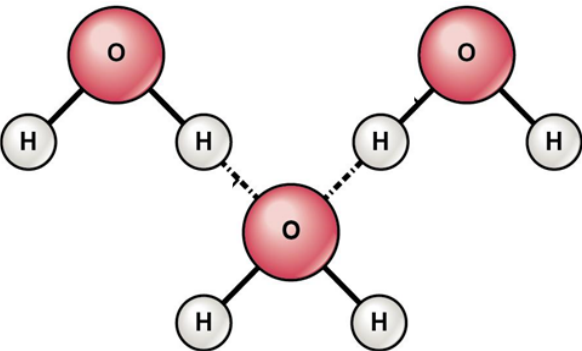
Experimental Group:

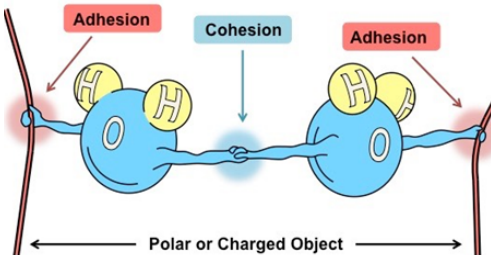
Independent Variable:

Dependent Variable:

Constant(s):

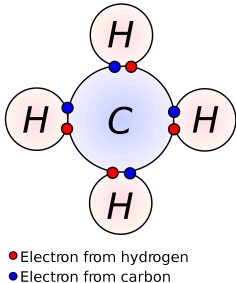
PROPERTIES OF WATER

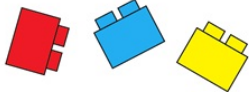
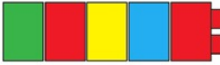
Water: The Molecule of Life	Hydrophilic vs. Hydrophobic
	<u>hydrophilic</u>- Example(s) of hydrophilic substances: <u>hydrophobic</u>- Example(s) of hydrophobic substances: 
Water is POLAR	
	Label which atom(s) have a partially negative and which atom(s) have a partially positive charge below:  Label the covalent and hydrogen bonds below: 

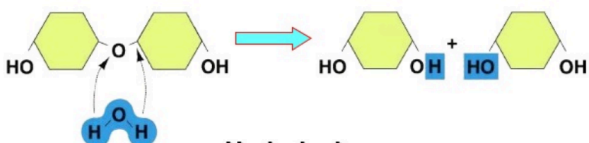
Property 1: Capillary Action	Property 2: Surface Tension
Example(s): 	Example(s):
Property 3: High Specific Heat	Property 4: Less Dense as a Solid
Example(s):	Example(s):
Property 5: Universal Solvent	Property 6: High Heat Of Vaporization
Example(s):	Example(s):

Describe what the pH scale is used for and general properties of acids and bases

TYPES OF REACTIONS & MACROMOLECULES

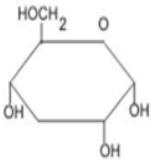
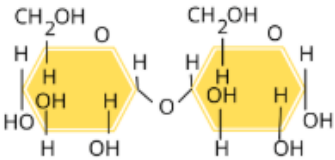
Organic Compounds	Inorganic compounds
Example(s):	Example(s):
Carbon Chemistry	
 • Electron from hydrogen • Electron from carbon	

Monomers vs. Polymers	
 Monomers	
 Polymer of three monomers	
 Polymer of five monomers	
Anabolic Reactions	Catabolic Reactions

Building Polymers: Dehydration Synthesis	Breaking Polymers: Hydrolysis
 Dehydration Synthesis	 Hydrolysis (Digestion)

List a few similarities and differences between dehydration synthesis and hydrolysis:

CARBOHYDRATES

Elements: Common identifiers: Building blocks: Main Function: Bond between monomers:	
	
Monosaccharides	Disaccharides
Examples: 	Examples: 

Polysaccharides	Polysaccharides in Plants
Polysaccharides in Animals	Polysaccharides in Other Organisms

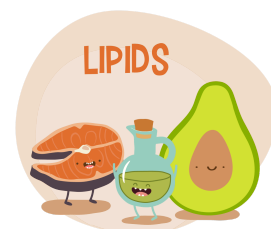
LIPIDS

Elements:

Common identifiers:

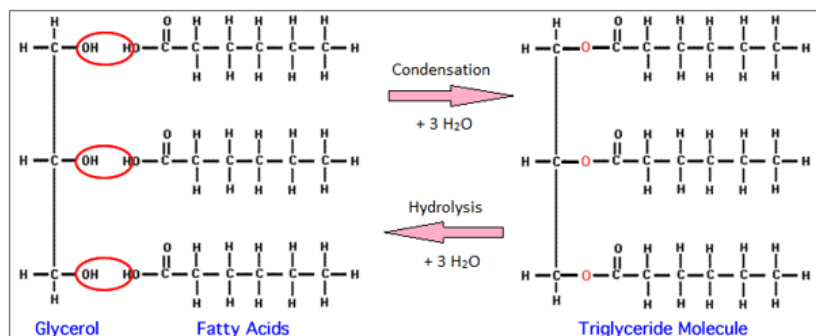
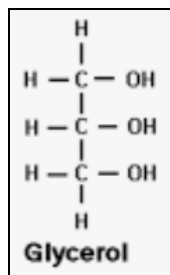
Bonds between monomers:

Examples:

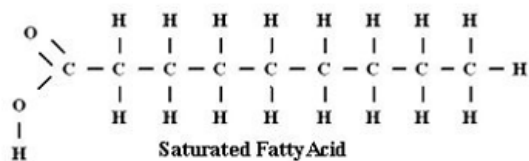


Lipids: Major Functions

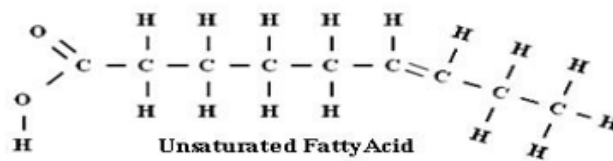
Lipids: Structure



Saturated Fatty Acids

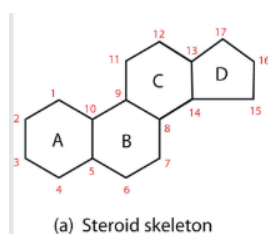


Unsaturated Fatty Acids

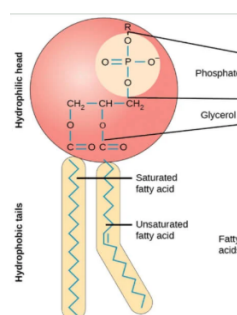


Why are saturated fats more commonly known as “bad fats”?

Steroids



Phospholipids



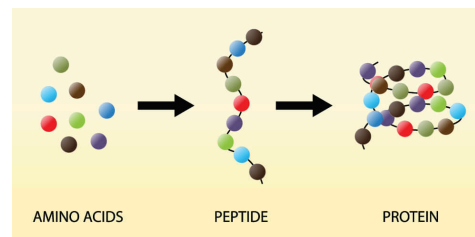
PROTEINS

Elements:

Building blocks:

Bond between monomers:

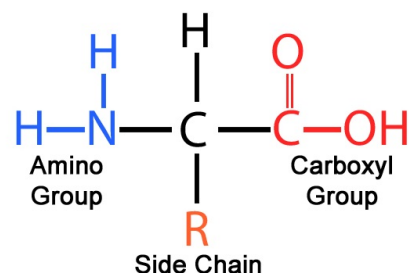
Polymer is called:



Amino Acid Structure

What do ALL amino acids contain?

What differs among amino acids and gives it unique characteristics?

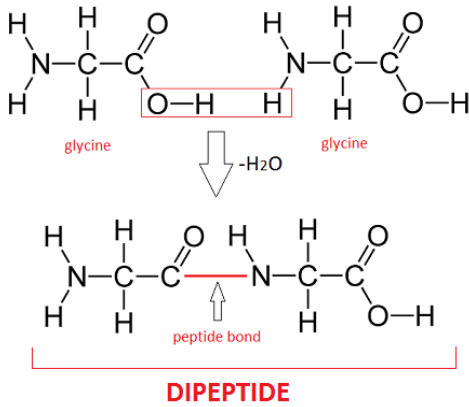


Proteins do ALL THE THINGS

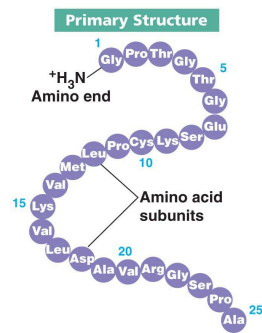
Role	Example	Function
Digestive Enzyme	Amylase, pepsin	
Transport	Hemoglobin	
Structure	Actin, keratin	
Hormone Signaling	Insulin	
Defense	Antibodies	
Contraction	Myosin	

CORE IDEA:

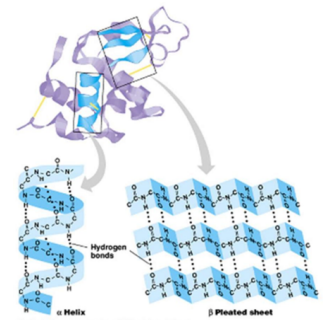
Protein Structure



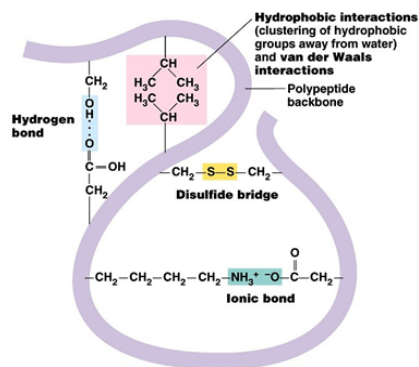
Primary Protein Structure



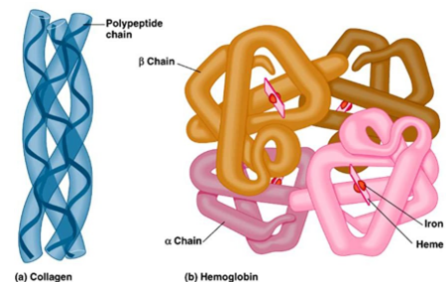
Secondary Protein Structure



Tertiary Protein Structure



Quaternary Protein Structure



NUCLEIC ACIDS

Elements:

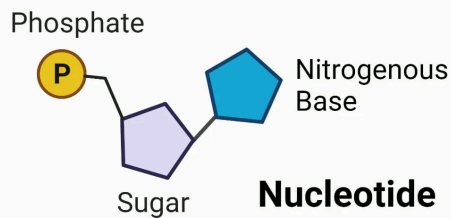
Building blocks:

Polymers (2):

Bonds:

Function:

Nucleotide Structure



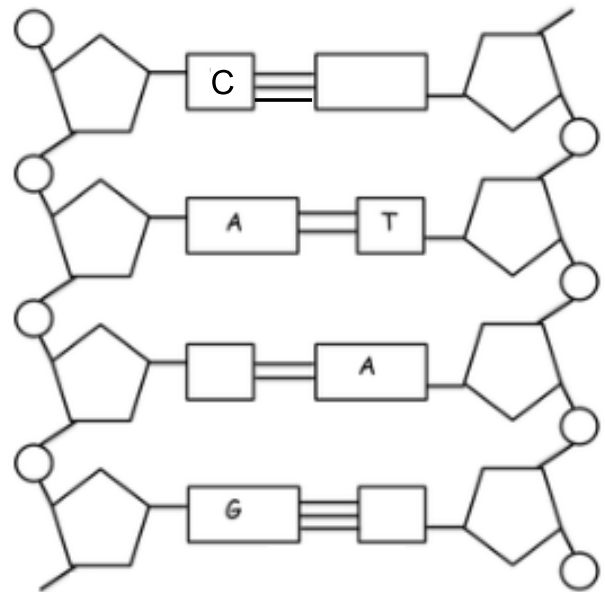
5 Nitrogenous Bases & Base Pairing

Nucleic Acid Structure

Label the missing base pairs in the diagram. Then, label at least one phosphodiester bond and one hydrogen bond.

Is this a strand of DNA or RNA? How do you know?

How many nucleotides are in this nucleic acid?



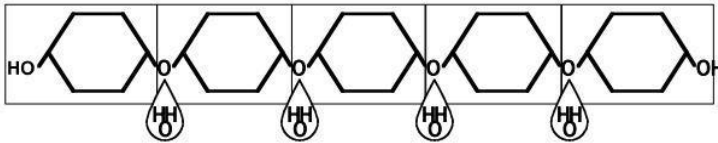
Identifying Categories of Biological Molecules

Dehydration Synthesis

Dehydration synthesis is a(n) _____ reaction. _____ is released as a byproduct when a _____ is released from one molecule and an _____ is released from another.

Draw two hexose structures remove a **H from one molecule and an OH from another**. Write H and OH on the side to represent your water droplet. Now draw the connected monomers together to form a bond. Include the water droplet below the bond to show that water is released as a byproduct. Repeat this until you have formed **3 bonds**.

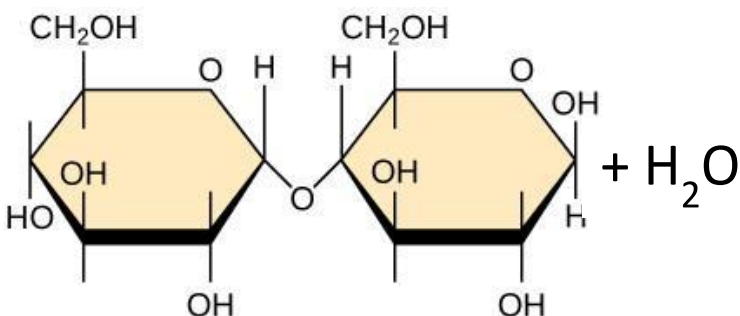
Example:



Hydrolysis

Hydrolysis is a(n) _____ reaction. This requires _____ to break the bonds. A H will bond to one molecule and the OH will bond to the other to create smaller molecules.

On the left, you will see two glucose molecules that have undergone dehydration synthesis and are now bonded together. On the right, draw two glucose molecules to show the bond being broken by water. Make sure to include the newly added H and OH.



Part 1: Macromolecules Overview

Organic molecules are the molecules which exist in all living things. They are life's building blocks. All things are formed from these organic molecules. There are four categories of organic molecules: carbohydrates, lipids, proteins, and nucleic acids.

1. How are organic molecules related to all living things?
2. Name four categories of organic molecules which form the basis of all living things:

Organic molecules have four common characteristics. First, they are all carbon based, meaning they all contain carbon. They are formed from just a few elements which join together to form small molecules which join together, or bond, to form large molecules. The third characteristic of all organic molecules is that each kind of organic molecule is built from a single type of building block called a **monomer**. For example, the monomer of carbohydrates is a monosaccharide (the sugars), the building block of lipids is a fatty acid and glycerol, the monomer of protein is an amino acid, and the monomer of a nucleic acid is the nucleotide. When these building blocks are joined together, they form a large molecule known as a **polymer**, just as bricks join together to form a wall. For example, sugars join together to form a carbohydrate.

3. All of the organic molecules are based on which element?
4. How are the building blocks of organic molecules like bricks?
5. What is the building block of each of the four classes of organic molecules?
 - a. The monomers of carbohydrates are:
 - b. The "monomers" of lipids are:
 - c. The monomer of proteins are:
6. What is a polymer?

The last common characteristic of all organic molecules is that their form determines their function. That means that their shape determines how they will behave and how they will react with other molecules. For example, the order of amino acids in a protein will determine the shape and function of the protein just as the order of words in a sentence shapes the meaning of the sentence.

7. What determines how organic molecules will look and behave?

PART 2: Carbohydrates

Sugars

Carbohydrates are the most common organic molecule because they make up most plant matter. They are made from carbon, hydrogen, and oxygen. Their building block, a **single sugar**, is called a **monosaccharide**. Sugars (also known as monosaccharides) consist of **carbon rings**. The names of sugars usually end in -ose (**sucrose**, **glucose**, **fructose**). Sugars are burned (oxidized) to release energy in cellular respiration and they play an important role in **homeostasis**. Glucose is the immediate source of energy for your cells.

Glue one example of a **MONOSACCHARIDE** in the space below.

Disaccharides

When two monosaccharides, or sugars, combine, they form a disaccharide (di=two). When two sugars undergo dehydration synthesis, the bond holding them together is called a **glycosidic linkage**.

Make and glue your **DISACCHARIDE** in the space below. Explain how these bonded together. What is the bond called that links them? Draw an arrow pointing to the bond and label it. Also show where water is leaving the bond during this reaction.

Starches

When more than two monosaccharides join together, a **polysaccharide** (poly=many) is formed. These include **starch**, **cellulose**, and **glycogen** (animal starch). **Starch** (in plants) and **glycogen** (in animals) are easily broken down into sugars for energy. **Cellulose**, on the other hand, which is made in plants, can be broken down only by a few organisms in the world (primarily the bacteria in the guts of termites). Yet all three types of macromolecules are made of long chains of sugar, and cellulose differs only by a small change in the connecting bond between each pair of sugars. **Make and glue a POLYSACCHARIDE in the space below.** Draw an arrow and label each bond that was formed when joining them. Also show where water is leaving the bond during this reaction.

1. What are the elements contained in carbohydrates?
2. What is the monomer of a carbohydrate?
3. What is a monosaccharide?
 - a. What does a monosaccharide look like?
 - b. What does the name usually end in?
4. What is a disaccharide?
5. How does a polysaccharide differ from a disaccharide?
6. What is the bond holding two sugars together?
7. Which polysaccharide involves food storage in **plants**?
8. Which polysaccharide involves food storage in **animals**?
 - a. Where is this usually stored?
9. What is cellulose used for?
10. Why would an athlete have a big pasta dinner the night before a race?

PART 3: Lipids

Lipids are a class of organic molecules which includes fats and oils, and has the function of a **long-term storage** (longer than carbohydrates) of energy in the body and protective insulation. The typical lipid structure is fatty acids plus glycerol. The fatty acids and glycerol are held together by an ester linkage (formed through dehydration synthesis).

1. What is the building block of a lipid?
2. What bond holds the glycerol and fatty acids together?
3. What is the difference between the function of carbohydrates and lipids and their importance in living organisms?

Fatty Acids

You should have some long **hydrocarbon chains** with a **carboxyl group** at one end. One of the defining features of these hydrocarbon chains is the absence of oxygen except in one carboxyl group at one end of the molecule. Fatty acids that contain no double bonds are **saturated**. An **unsaturated fatty acid** contains one or more double bonds.

Glue your FATTY ACIDS in the space below. Label each one as **saturated** or **unsaturated**. If it is unsaturated, **draw an arrow pointing to the part in the hydrocarbon chain where it is unsaturated.**

1. What is a saturated fatty acid?
2. How does an unsaturated fatty acid differ from a saturated fatty acid?
3. What types of foods contain high levels of saturated fats? Unsaturated?
4. Which type of fatty acid is considered “unhealthy” and why is it considered more unhealthy than the other?

Triglycerides

Lipids do not form true polymers. Instead, we say they are big molecules made up of smaller subunits. One of the big molecules is a triglyceride. This is made from **one glycerol plus three fatty acids**.

Glue the TRIGLYCERIDE in the space below. Draw an arrow pointing to each place where a fatty acid was bonded to the glycerol molecule. Label this specific bond. In addition, identify if the molecule is saturated or unsaturated.

Phospholipids

Another big lipid molecule is the phospholipid. Phospholipids are composed of one phosphate group, one glycerol, and two fatty acid tails. Phospholipids compose the cell membrane. Part of the phospholipid is hydrophilic while the other part is hydrophobic. This structure is fundamental to the cell membrane functioning as a selectively permeable membrane.

Glue the PHOSPHOLIPID in the space below. Draw an arrow pointing to the hydrophilic portion and an arrow pointing to the hydrophobic portion.

1. What is the difference between a phospholipid and a triglyceride?
2. What bond is found in both phospholipids and triglycerides?

Steroids

Steroids are another lipid. The steroid cholesterol plays an important role in strengthening the cell membrane. Other steroids act as hormones. Steroids can be recognized by their **multiple rings of carbon atoms fused together**.

Glue two examples of steroids below.

PART 4: Proteins

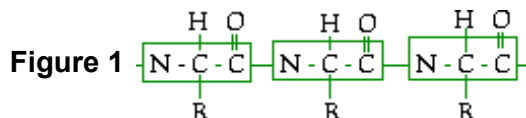
Amino Acids

Amino acids are the monomers of proteins - molecules that play many important roles in the body (including muscle structure, hormones, transport proteins for carrying molecules across cell membranes, and enzymes). Each amino acid contains a central carbon, a hydrogen, a carboxyl group, an amino group, and an R group. Each amino acid has a different side group that is represented by "R." The R-groups can be hydrophilic or hydrophobic, which plays a role in how the protein folds.

1. What are some of the functions of proteins?
2. What is the monomer of proteins?
3. What is the name of the bond that joins amino acids?

In the space below, find and glue **two DIFFERENT amino acids**. Label the **N-Terminus**, **C-Terminus**, and **R-Group**.

Polypeptides are the polymers of proteins. The amino acids are held together by **peptide bonds** to form a long chain (polypeptide). The "-N-C-C-" in the center of the molecule is known as its backbone. This is shown below in Figure 1.



Make a polypeptide that is **THREE amino acids long**. Label the **peptide bonds**, **N-Terminus**, and **C Terminus**.

There are **20** different kinds of amino acids. The only difference in the 20 kinds of amino acids is the "R" group. Some "R" groups are very small, others are large, some are hydrophilic, some are hydrophobic, and others form chains and rings. The sequence and shapes of the "R" groups control the shape and function of the protein.

1. How many different amino acids are there?
2. What part of the amino acid varies from one amino acid to another?
3. What determines the shape and function of a protein?

PART 5: Nucleic Acids

Nucleotides

Nucleotides are the monomers of nucleic acids. They are made of three components: a phosphate group, a 5-carbon sugar, and a nitrogenous base. There are 5 nitrogenous bases: adenine, guanine, cytosine, thymine (only found in DNA), and uracil (only found in RNA).

1. What is the function of nucleic acids?
2. In the space below, **draw a nucleotide and label the parts**.